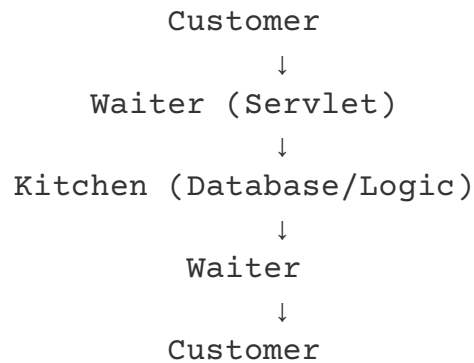


EXAM LAST MIN REVISION UNIT 4

WHAT IS SERVLET

Easy Analogy

Think of a restaurant:



The waiter receives the order, talks to the kitchen, and brings back the result.

Similarly:



Exam Points

- Servlet is a server-side Java program.
- Used to create dynamic web applications.
- Runs inside a web container such as Apache Tomcat.
- Handles client requests and responses.
- Usually extends HttpServlet.
- Part of the Controller layer in MVC architecture.

- Faster and more efficient than CGI.

Quick Revision (2 Marks)

A Servlet is a server-side Java program that processes client requests and generates dynamic responses. It runs inside a web container such as Tomcat and is mainly used as the Controller component in MVC-based web applications.

SERVLET LIFECYCLE

Servlet vs JSP Life Cycle

JSP	Servlet
jspInit()	init()
_jspService()	service()
jspDestroy()	destroy()

Remember:

JSP
↓
Converted to Servlet

So their life cycles are very similar.

Exam Points

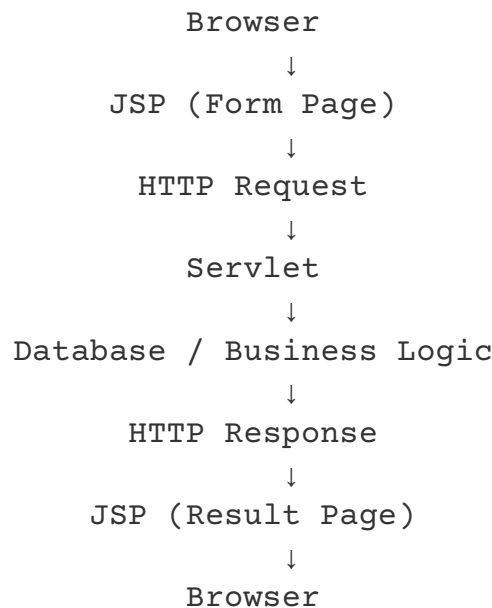
- Life cycle is managed by the container.
- Four stages: Loading, Initialization, Request Processing, Destruction.
- init() runs once.
- service() runs for every request.
- destroy() runs once.
- doGet() handles GET requests.
- doPost() handles POST requests.

Quick Revision (2 Marks)

The Servlet Life Cycle consists of Loading, Initialization, Request Processing, and Destruction. The container calls `init()` once, `service()` for every request, and `destroy()` once before removing the Servlet from memory.

HANDLING HTTP REQUESTS AND RESPONSES

Complete Flow of HTTP REQ (BROWSER -> SERVER)



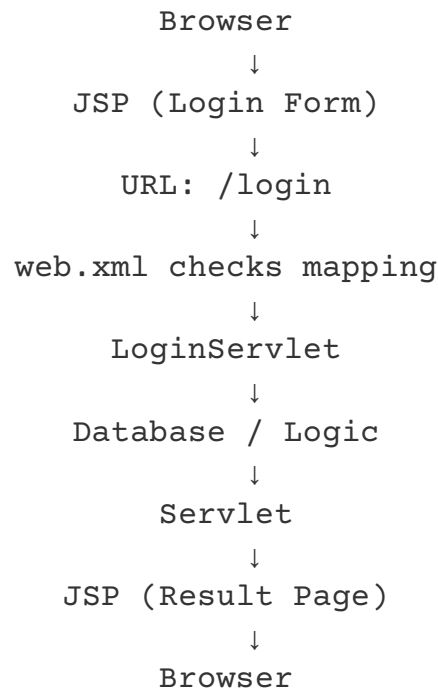
Exam Points

- Request is sent from browser to server.
- Response is sent from server to browser.
- `HttpServletRequest` handles requests.
- `HttpServletResponse` handles responses.
- `doGet()` handles GET requests.
- `doPost()` handles POST requests.
- Request receives data.
- Response sends data.

Quick Revision (2 Marks)

HTTP Request and Response are used for communication between the browser and Servlet. `HttpServletRequest` receives client data, while `HttpServletResponse` sends data back to the client. GET is mainly used for retrieving data and POST is mainly used for sending data securely.

USE CASES OF WEB.XML FILE



web.xml vs Annotation

web.xml	Annotation
XML based	Java based
Centralized configuration	Configuration inside code
Common in old projects	Common in modern projects
Easy to manage large apps	Easy for small apps

Important Uses of web.xml

- Servlet Mapping
- Welcome Pages
- Error Pages
- Session Configuration
- Initialization Parameters
- Filter Configuration
- Security Configuration

Exam Points

- Deployment Descriptor file name is web.xml.
- Located inside WEB-INF.
- Used to configure web applications.

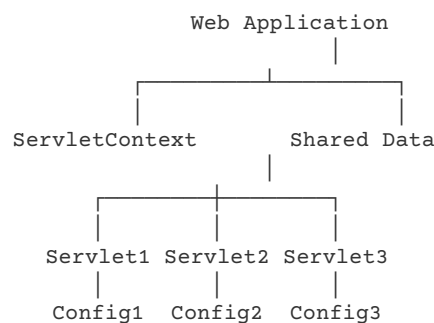
- Maps URLs to Servlets.
- Configures welcome pages and error pages.
- Stores initialization parameters.
- Read by Tomcat during application startup.
- Can be replaced by annotations in modern applications.

Quick Revision (2 Marks)

The Deployment Descriptor (web.xml) is an XML configuration file used to configure Java web applications. It defines Servlet mappings, welcome pages, error pages, session settings, and initialization parameters. It is located inside the WEB-INF directory and is read by the web container during startup.

WHAT IS SERVLETCONTEXT?

Complete Comparison Diagram



ServletContext	ServletConfig
One per application	One per Servlet
Shared by all Servlets	Specific to one Servlet
Application-wide data	Servlet-specific data
Created once for application	Created for each Servlet
Global scope	Local scope

Exam Points

- ServletContext is application-wide.
- ServletConfig is Servlet-specific.
- One ServletContext per application.
- One ServletConfig per Servlet.

- ServletContext shares data among Servlets.
- ServletConfig stores initialization parameters.
- Both are created by the container.

Quick Revision (2 Marks)

ServletContext is a global object shared by all Servlets in a web application and is used to store application-wide data. ServletConfig is a Servlet-specific object used to store configuration parameters for an individual Servlet. One ServletContext exists per application, while each Servlet has its own ServletConfig.

SESSION MANAGEMENT AND COOKIES

Cookie	Session
Stored in Browser	Stored on Server
Less Secure	More Secure
Small Data	Larger Data
User Can Delete	Controlled by Server
Faster Access	More Server Memory Used

Why Session is Needed?

Without Session:

Login Page → Profile Page → Server Forgets User (Problem )

With Session:

Login Page → Session Stores User → Profile Page → Cart Page → Orders Page

Easy Memory Trick: Cookie → Browser Memory | Session → Server Memory

Exam Points

- HTTP is stateless.
- Session Management remembers users.
- Cookies are stored in browser.
- Sessions are stored on server.

- HttpSession is used for user tracking.
- session.invalidate() destroys session.
- Cookies store small pieces of information.
- Session is more secure than Cookies.

Quick Revision (2 Marks)

Session Management is used to maintain user information across multiple requests because HTTP is stateless. Cookies store data in the browser, while HttpSession stores data on the server. Sessions are commonly used for login management, shopping carts, and user tracking.

SERVLET CHAINING & FILTERS

Easy Analogy

Servlet Chaining: Think of a factory. Step 1 → Step 2 → Step 3. Multiple workers complete the product.

Filter: Think of a security guard. Visitor → Security Guard → Office. The guard checks everyone before entry.

Comparison

Servlet Chaining	Filter
Multiple Servlets work together	Intercepts requests/responses
Request moves between Servlets	Request passes through Filter
Used for task division	Used for common processing
Implemented using forward/include	Implemented using Filter interface

Exam Points

- **Servlet Chaining:** One Servlet forwards request to another; Uses RequestDispatcher; Can use forward() or include().
- **Filters:** Intercept requests and responses; Implement Filter interface; Commonly used for authentication, logging, and security; chain.doFilter() passes request forward.

Quick Revision (2 Marks)

Servlet Chaining allows multiple Servlets to process the same request using `forward()` and `include()`. Filters are special components that intercept requests and responses before they reach the Servlet and are commonly used for authentication, logging, and validation.

FILE UPLOAD AND DOWNLOAD IN SERVLET

Upload vs Download

Upload	Download
Browser → Server	Server → Browser
User sends file	User receives file
Uses multipart/form-data	Uses response output stream
Stores file on server	Retrieves file from server

Exam Points

- **File Upload:** Uses multipart/form-data; Uses `@MultipartConfig`; Uses `request.getPart()`; Stores files on server.
- **File Download:** Uses `HttpServletResponse`; Uses `setContentType()`; Uses `getOutputStream()`; Sends files to browser.

Quick Revision (2 Marks)

File Upload allows users to send files from the browser to the server using multipart/form-data and `@MultipartConfig`. File Download allows users to retrieve files from the server using `HttpServletResponse`, content types, and output streams.